Model Architecture: gpt-oss-20b

Generated by sg_modelload

July 21, 2026

1 Model Overview

Model Name	gpt-oss-20b
Path	/models/gpt-oss-20b
Total Parameters	20,914,757,184
Weight Dtype	bfloat16

2 Forward Pass Algorithm

Algorithm 1 Forward pass of gpt-oss-20b

$h \leftarrow \text{embed_tokens}(x) \in \mathbb{R}^{B \times T \times 2880}$
for $layer \leftarrow 0$ **to** 23 **do**
    $h \leftarrow \text{layer}_{layer}(h)$
end for
$h \leftarrow \text{norm}(h) \in \mathbb{R}^{B \times T \times 2880}$
$\text{logits} \leftarrow \text{lm_head}(h) \in \mathbb{R}^{B \times T \times 201088}$

3 Layer-by-Layer Architecture

All weight tensors and their dimensions are listed below.

Layer Path	Type	Weight Shape	Dtype	Params
model.layers.0	GptOssDecoderLayer	–	–	–
model.layers.0.input_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.0.mlp	GptOssMLP	–	–	–
model.layers.0.mlp.experts.gate_up_proj	GptOssExperts	32 × 2880 × 5760	bfloat16	530,841,600
model.layers.0.mlp.experts.gate_up_proj_bias	GptOssExperts	32 × 5760	bfloat16	184,320
model.layers.0.mlp.experts.down_proj	GptOssExperts	32 × 2880 × 2880	bfloat16	265,420,800
model.layers.0.mlp.experts.down_proj_bias	GptOssExperts	32 × 2880	bfloat16	92,160
model.layers.0.mlp.router.weight	GptOssTopKRouter	32 × 2880	bfloat16	92,160
model.layers.0.mlp.router.bias	GptOssTopKRouter	32	bfloat16	32
model.layers.0.post_attention_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.0.self_attn.sinks	GptOssAttention	64	bfloat16	64
model.layers.0.self_attn.k_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.0.self_attn.k_proj.bias	Linear	512	bfloat16	512
model.layers.0.self_attn.o_proj.weight	Linear	2880 × 4096	bfloat16	11,796,480
model.layers.0.self_attn.o_proj.bias	Linear	2880	bfloat16	2,880
model.layers.0.self_attn.q_proj.weight	Linear	4096 × 2880	bfloat16	11,796,480
model.layers.0.self_attn.q_proj.bias	Linear	4096	bfloat16	4,096
model.layers.0.self_attn.v_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.0.self_attn.v_proj.bias	Linear	512	bfloat16	512
model.layers.1	GptOssDecoderLayer	–	–	–
model.layers.1.input_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.1.mlp	GptOssMLP	–	–	–
model.layers.1.mlp.experts.gate_up_proj	GptOssExperts	32 × 2880 × 5760	bfloat16	530,841,600

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Layer Path	Type	Weight Shape	Dtype	Params
model.layers.1.mlp.experts.gate_up_proj_bias	GptOssExperts	32 × 5760	bfloat16	184,320
model.layers.1.mlp.experts.down_proj	GptOssExperts	32 × 2880 × 2880	bfloat16	265,420,800
model.layers.1.mlp.experts.down_proj_bias	GptOssExperts	32 × 2880	bfloat16	92,160
model.layers.1.mlp.router.weight	GptOssTopKRouter	32 × 2880	bfloat16	92,160
model.layers.1.mlp.router.bias	GptOssTopKRouter	32	bfloat16	32
model.layers.1.post_attention_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.1.self_attn.sinks	GptOssAttention	64	bfloat16	64
model.layers.1.self_attn.k_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.1.self_attn.k_proj.bias	Linear	512	bfloat16	512
model.layers.1.self_attn.o_proj.weight	Linear	2880 × 4096	bfloat16	11,796,480
model.layers.1.self_attn.o_proj.bias	Linear	2880	bfloat16	2,880
model.layers.1.self_attn.q_proj.weight	Linear	4096 × 2880	bfloat16	11,796,480
model.layers.1.self_attn.q_proj.bias	Linear	4096	bfloat16	4,096
model.layers.1.self_attn.v_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.1.self_attn.v_proj.bias	Linear	512	bfloat16	512
model.layers.2	GptOssDecoderLayer	–	–	–
model.layers.2.input_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.2.mlp	GptOssMLP	–	–	–
model.layers.2.mlp.experts.gate_up_proj	GptOssExperts	32 × 2880 × 5760	bfloat16	530,841,600
model.layers.2.mlp.experts.gate_up_proj_bias	GptOssExperts	32 × 5760	bfloat16	184,320
model.layers.2.mlp.experts.down_proj	GptOssExperts	32 × 2880 × 2880	bfloat16	265,420,800
model.layers.2.mlp.experts.down_proj_bias	GptOssExperts	32 × 2880	bfloat16	92,160
model.layers.2.mlp.router.weight	GptOssTopKRouter	32 × 2880	bfloat16	92,160
model.layers.2.mlp.router.bias	GptOssTopKRouter	32	bfloat16	32
model.layers.2.post_attention_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.2.self_attn.sinks	GptOssAttention	64	bfloat16	64
model.layers.2.self_attn.k_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.2.self_attn.k_proj.bias	Linear	512	bfloat16	512
model.layers.2.self_attn.o_proj.weight	Linear	2880 × 4096	bfloat16	11,796,480
model.layers.2.self_attn.o_proj.bias	Linear	2880	bfloat16	2,880
model.layers.2.self_attn.q_proj.weight	Linear	4096 × 2880	bfloat16	11,796,480
model.layers.2.self_attn.q_proj.bias	Linear	4096	bfloat16	4,096
model.layers.2.self_attn.v_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.2.self_attn.v_proj.bias	Linear	512	bfloat16	512
model.layers.3	GptOssDecoderLayer	–	–	–
model.layers.3.input_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.3.mlp	GptOssMLP	–	–	–
model.layers.3.mlp.experts.gate_up_proj	GptOssExperts	32 × 2880 × 5760	bfloat16	530,841,600
model.layers.3.mlp.experts.gate_up_proj_bias	GptOssExperts	32 × 5760	bfloat16	184,320
model.layers.3.mlp.experts.down_proj	GptOssExperts	32 × 2880 × 2880	bfloat16	265,420,800
model.layers.3.mlp.experts.down_proj_bias	GptOssExperts	32 × 2880	bfloat16	92,160
model.layers.3.mlp.router.weight	GptOssTopKRouter	32 × 2880	bfloat16	92,160
model.layers.3.mlp.router.bias	GptOssTopKRouter	32	bfloat16	32
model.layers.3.post_attention_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.3.self_attn.sinks	GptOssAttention	64	bfloat16	64
model.layers.3.self_attn.k_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.3.self_attn.k_proj.bias	Linear	512	bfloat16	512
model.layers.3.self_attn.o_proj.weight	Linear	2880 × 4096	bfloat16	11,796,480
model.layers.3.self_attn.o_proj.bias	Linear	2880	bfloat16	2,880
model.layers.3.self_attn.q_proj.weight	Linear	4096 × 2880	bfloat16	11,796,480
model.layers.3.self_attn.q_proj.bias	Linear	4096	bfloat16	4,096
model.layers.3.self_attn.v_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.3.self_attn.v_proj.bias	Linear	512	bfloat16	512
model.layers.4	GptOssDecoderLayer	–	–	–
model.layers.4.input_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.4.mlp	GptOssMLP	–	–	–
model.layers.4.mlp.experts.gate_up_proj	GptOssExperts	32 × 2880 × 5760	bfloat16	530,841,600
model.layers.4.mlp.experts.gate_up_proj_bias	GptOssExperts	32 × 5760	bfloat16	184,320

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Layer Path	Type	Weight Shape	Dtype	Params
model.layers.4.mlp.experts.down_proj	GptOssExperts	32 × 2880 × 2880	bfloat16	265,420,800
model.layers.4.mlp.experts.down_proj_bias	GptOssExperts	32 × 2880	bfloat16	92,160
model.layers.4.mlp.router.weight	GptOssTopKRouter	32 × 2880	bfloat16	92,160
model.layers.4.mlp.router.bias	GptOssTopKRouter	32	bfloat16	32
model.layers.4.post_attention_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.4.self_attn.sinks	GptOssAttention	64	bfloat16	64
model.layers.4.self_attn.k_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.4.self_attn.k_proj.bias	Linear	512	bfloat16	512
model.layers.4.self_attn.o_proj.weight	Linear	2880 × 4096	bfloat16	11,796,480
model.layers.4.self_attn.o_proj.bias	Linear	2880	bfloat16	2,880
model.layers.4.self_attn.q_proj.weight	Linear	4096 × 2880	bfloat16	11,796,480
model.layers.4.self_attn.q_proj.bias	Linear	4096	bfloat16	4,096
model.layers.4.self_attn.v_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.4.self_attn.v_proj.bias	Linear	512	bfloat16	512
model.layers.5	GptOssDecoderLayer	–	–	–
model.layers.5.input_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.5.mlp	GptOssMLP	–	–	–
model.layers.5.mlp.experts.gate_up_proj	GptOssExperts	32 × 2880 × 5760	bfloat16	530,841,600
model.layers.5.mlp.experts.gate_up_proj_bias	GptOssExperts	32 × 5760	bfloat16	184,320
model.layers.5.mlp.experts.down_proj	GptOssExperts	32 × 2880 × 2880	bfloat16	265,420,800
model.layers.5.mlp.experts.down_proj_bias	GptOssExperts	32 × 2880	bfloat16	92,160
model.layers.5.mlp.router.weight	GptOssTopKRouter	32 × 2880	bfloat16	92,160
model.layers.5.mlp.router.bias	GptOssTopKRouter	32	bfloat16	32
model.layers.5.post_attention_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.5.self_attn.sinks	GptOssAttention	64	bfloat16	64
model.layers.5.self_attn.k_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.5.self_attn.k_proj.bias	Linear	512	bfloat16	512
model.layers.5.self_attn.o_proj.weight	Linear	2880 × 4096	bfloat16	11,796,480
model.layers.5.self_attn.o_proj.bias	Linear	2880	bfloat16	2,880
model.layers.5.self_attn.q_proj.weight	Linear	4096 × 2880	bfloat16	11,796,480
model.layers.5.self_attn.q_proj.bias	Linear	4096	bfloat16	4,096
model.layers.5.self_attn.v_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.5.self_attn.v_proj.bias	Linear	512	bfloat16	512
model.layers.6	GptOssDecoderLayer	–	–	–
model.layers.6.input_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.6.mlp	GptOssMLP	–	–	–
model.layers.6.mlp.experts.gate_up_proj	GptOssExperts	32 × 2880 × 5760	bfloat16	530,841,600
model.layers.6.mlp.experts.gate_up_proj_bias	GptOssExperts	32 × 5760	bfloat16	184,320
model.layers.6.mlp.experts.down_proj	GptOssExperts	32 × 2880 × 2880	bfloat16	265,420,800
model.layers.6.mlp.experts.down_proj_bias	GptOssExperts	32 × 2880	bfloat16	92,160
model.layers.6.mlp.router.weight	GptOssTopKRouter	32 × 2880	bfloat16	92,160
model.layers.6.mlp.router.bias	GptOssTopKRouter	32	bfloat16	32
model.layers.6.post_attention_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.6.self_attn.sinks	GptOssAttention	64	bfloat16	64
model.layers.6.self_attn.k_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.6.self_attn.k_proj.bias	Linear	512	bfloat16	512
model.layers.6.self_attn.o_proj.weight	Linear	2880 × 4096	bfloat16	11,796,480
model.layers.6.self_attn.o_proj.bias	Linear	2880	bfloat16	2,880
model.layers.6.self_attn.q_proj.weight	Linear	4096 × 2880	bfloat16	11,796,480
model.layers.6.self_attn.q_proj.bias	Linear	4096	bfloat16	4,096
model.layers.6.self_attn.v_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.6.self_attn.v_proj.bias	Linear	512	bfloat16	512
model.layers.7	GptOssDecoderLayer	–	–	–
model.layers.7.input_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.7.mlp	GptOssMLP	–	–	–
model.layers.7.mlp.experts.gate_up_proj	GptOssExperts	32 × 2880 × 5760	bfloat16	530,841,600
model.layers.7.mlp.experts.gate_up_proj_bias	GptOssExperts	32 × 5760	bfloat16	184,320
model.layers.7.mlp.experts.down_proj	GptOssExperts	32 × 2880 × 2880	bfloat16	265,420,800

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Layer Path	Type	Weight Shape	Dtype	Params
model.layers.7.mlp.experts.down_proj_bias	GptOssExperts	32 × 2880	bfloat16	92,160
model.layers.7.mlp.router.weight	GptOssTopKRouter	32 × 2880	bfloat16	92,160
model.layers.7.mlp.router.bias	GptOssTopKRouter	32	bfloat16	32
model.layers.7.post_attention_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.7.self_attn.sinks	GptOssAttention	64	bfloat16	64
model.layers.7.self_attn.k_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.7.self_attn.k_proj.bias	Linear	512	bfloat16	512
model.layers.7.self_attn.o_proj.weight	Linear	2880 × 4096	bfloat16	11,796,480
model.layers.7.self_attn.o_proj.bias	Linear	2880	bfloat16	2,880
model.layers.7.self_attn.q_proj.weight	Linear	4096 × 2880	bfloat16	11,796,480
model.layers.7.self_attn.q_proj.bias	Linear	4096	bfloat16	4,096
model.layers.7.self_attn.v_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.7.self_attn.v_proj.bias	Linear	512	bfloat16	512
model.layers.8	GptOssDecoderLayer	–	–	–
model.layers.8.input_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.8.mlp	GptOssMLP	–	–	–
model.layers.8.mlp.experts.gate_up_proj	GptOssExperts	32 × 2880 × 5760	bfloat16	530,841,600
model.layers.8.mlp.experts.gate_up_proj_bias	GptOssExperts	32 × 5760	bfloat16	184,320
model.layers.8.mlp.experts.down_proj	GptOssExperts	32 × 2880 × 2880	bfloat16	265,420,800
model.layers.8.mlp.experts.down_proj_bias	GptOssExperts	32 × 2880	bfloat16	92,160
model.layers.8.mlp.router.weight	GptOssTopKRouter	32 × 2880	bfloat16	92,160
model.layers.8.mlp.router.bias	GptOssTopKRouter	32	bfloat16	32
model.layers.8.post_attention_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.8.self_attn.sinks	GptOssAttention	64	bfloat16	64
model.layers.8.self_attn.k_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.8.self_attn.k_proj.bias	Linear	512	bfloat16	512
model.layers.8.self_attn.o_proj.weight	Linear	2880 × 4096	bfloat16	11,796,480
model.layers.8.self_attn.o_proj.bias	Linear	2880	bfloat16	2,880
model.layers.8.self_attn.q_proj.weight	Linear	4096 × 2880	bfloat16	11,796,480
model.layers.8.self_attn.q_proj.bias	Linear	4096	bfloat16	4,096
model.layers.8.self_attn.v_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.8.self_attn.v_proj.bias	Linear	512	bfloat16	512
model.layers.9	GptOssDecoderLayer	–	–	–
model.layers.9.input_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.9.mlp	GptOssMLP	–	–	–
model.layers.9.mlp.experts.gate_up_proj	GptOssExperts	32 × 2880 × 5760	bfloat16	530,841,600
model.layers.9.mlp.experts.gate_up_proj_bias	GptOssExperts	32 × 5760	bfloat16	184,320
model.layers.9.mlp.experts.down_proj	GptOssExperts	32 × 2880 × 2880	bfloat16	265,420,800
model.layers.9.mlp.experts.down_proj_bias	GptOssExperts	32 × 2880	bfloat16	92,160
model.layers.9.mlp.router.weight	GptOssTopKRouter	32 × 2880	bfloat16	92,160
model.layers.9.mlp.router.bias	GptOssTopKRouter	32	bfloat16	32
model.layers.9.post_attention_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.9.self_attn.sinks	GptOssAttention	64	bfloat16	64
model.layers.9.self_attn.k_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.9.self_attn.k_proj.bias	Linear	512	bfloat16	512
model.layers.9.self_attn.o_proj.weight	Linear	2880 × 4096	bfloat16	11,796,480
model.layers.9.self_attn.o_proj.bias	Linear	2880	bfloat16	2,880
model.layers.9.self_attn.q_proj.weight	Linear	4096 × 2880	bfloat16	11,796,480
model.layers.9.self_attn.q_proj.bias	Linear	4096	bfloat16	4,096
model.layers.9.self_attn.v_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.9.self_attn.v_proj.bias	Linear	512	bfloat16	512
model.layers.10	GptOssDecoderLayer	–	–	–
model.layers.10.input_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.10.mlp	GptOssMLP	–	–	–
model.layers.10.mlp.experts.gate_up_proj	GptOssExperts	32 × 2880 × 5760	bfloat16	530,841,600
model.layers.10.mlp.experts.gate_up_proj_bias	GptOssExperts	32 × 5760	bfloat16	184,320
model.layers.10.mlp.experts.down_proj	GptOssExperts	32 × 2880 × 2880	bfloat16	265,420,800
model.layers.10.mlp.experts.down_proj_bias	GptOssExperts	32 × 2880	bfloat16	92,160

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Layer Path	Type	Weight Shape	Dtype	Params
model.layers.10.mlp.router.weight	GptOssTopKRouter	32 × 2880	bfloat16	92,160
model.layers.10.mlp.router.bias	GptOssTopKRouter	32	bfloat16	32
model.layers.10.post_attention_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.10.self_attn.sinks	GptOssAttention	64	bfloat16	64
model.layers.10.self_attn.k_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.10.self_attn.k_proj.bias	Linear	512	bfloat16	512
model.layers.10.self_attn.o_proj.weight	Linear	2880 × 4096	bfloat16	11,796,480
model.layers.10.self_attn.o_proj.bias	Linear	2880	bfloat16	2,880
model.layers.10.self_attn.q_proj.weight	Linear	4096 × 2880	bfloat16	11,796,480
model.layers.10.self_attn.q_proj.bias	Linear	4096	bfloat16	4,096
model.layers.10.self_attn.v_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.10.self_attn.v_proj.bias	Linear	512	bfloat16	512
model.layers.11	GptOssDecoderLayer	–	–	–
model.layers.11.input_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.11.mlp	GptOssMLP	–	–	–
model.layers.11.mlp.experts.gate_up_proj	GptOssExperts	32 × 2880 × 5760	bfloat16	530,841,600
model.layers.11.mlp.experts.gate_up_proj_bias	GptOssExperts	32 × 5760	bfloat16	184,320
model.layers.11.mlp.experts.down_proj	GptOssExperts	32 × 2880 × 2880	bfloat16	265,420,800
model.layers.11.mlp.experts.down_proj_bias	GptOssExperts	32 × 2880	bfloat16	92,160
model.layers.11.mlp.router.weight	GptOssTopKRouter	32 × 2880	bfloat16	92,160
model.layers.11.mlp.router.bias	GptOssTopKRouter	32	bfloat16	32
model.layers.11.post_attention_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.11.self_attn.sinks	GptOssAttention	64	bfloat16	64
model.layers.11.self_attn.k_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.11.self_attn.k_proj.bias	Linear	512	bfloat16	512
model.layers.11.self_attn.o_proj.weight	Linear	2880 × 4096	bfloat16	11,796,480
model.layers.11.self_attn.o_proj.bias	Linear	2880	bfloat16	2,880
model.layers.11.self_attn.q_proj.weight	Linear	4096 × 2880	bfloat16	11,796,480
model.layers.11.self_attn.q_proj.bias	Linear	4096	bfloat16	4,096
model.layers.11.self_attn.v_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.11.self_attn.v_proj.bias	Linear	512	bfloat16	512
model.layers.12	GptOssDecoderLayer	–	–	–
model.layers.12.input_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.12.mlp	GptOssMLP	–	–	–
model.layers.12.mlp.experts.gate_up_proj	GptOssExperts	32 × 2880 × 5760	bfloat16	530,841,600
model.layers.12.mlp.experts.gate_up_proj_bias	GptOssExperts	32 × 5760	bfloat16	184,320
model.layers.12.mlp.experts.down_proj	GptOssExperts	32 × 2880 × 2880	bfloat16	265,420,800
model.layers.12.mlp.experts.down_proj_bias	GptOssExperts	32 × 2880	bfloat16	92,160
model.layers.12.mlp.router.weight	GptOssTopKRouter	32 × 2880	bfloat16	92,160
model.layers.12.mlp.router.bias	GptOssTopKRouter	32	bfloat16	32
model.layers.12.post_attention_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.12.self_attn.sinks	GptOssAttention	64	bfloat16	64
model.layers.12.self_attn.k_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.12.self_attn.k_proj.bias	Linear	512	bfloat16	512
model.layers.12.self_attn.o_proj.weight	Linear	2880 × 4096	bfloat16	11,796,480
model.layers.12.self_attn.o_proj.bias	Linear	2880	bfloat16	2,880
model.layers.12.self_attn.q_proj.weight	Linear	4096 × 2880	bfloat16	11,796,480
model.layers.12.self_attn.q_proj.bias	Linear	4096	bfloat16	4,096
model.layers.12.self_attn.v_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.12.self_attn.v_proj.bias	Linear	512	bfloat16	512
model.layers.13	GptOssDecoderLayer	–	–	–
model.layers.13.input_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.13.mlp	GptOssMLP	–	–	–
model.layers.13.mlp.experts.gate_up_proj	GptOssExperts	32 × 2880 × 5760	bfloat16	530,841,600
model.layers.13.mlp.experts.gate_up_proj_bias	GptOssExperts	32 × 5760	bfloat16	184,320
model.layers.13.mlp.experts.down_proj	GptOssExperts	32 × 2880 × 2880	bfloat16	265,420,800
model.layers.13.mlp.experts.down_proj_bias	GptOssExperts	32 × 2880	bfloat16	92,160
model.layers.13.mlp.router.weight	GptOssTopKRouter	32 × 2880	bfloat16	92,160

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Layer Path	Type	Weight Shape	Dtype	Params
model.layers.13.mlp.router.bias	GptOssTopKRouter	32	bfloat16	32
model.layers.13.post_attention_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.13.self_attn.sinks	GptOssAttention	64	bfloat16	64
model.layers.13.self_attn.k_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.13.self_attn.k_proj.bias	Linear	512	bfloat16	512
model.layers.13.self_attn.o_proj.weight	Linear	2880 × 4096	bfloat16	11,796,480
model.layers.13.self_attn.o_proj.bias	Linear	2880	bfloat16	2,880
model.layers.13.self_attn.q_proj.weight	Linear	4096 × 2880	bfloat16	11,796,480
model.layers.13.self_attn.q_proj.bias	Linear	4096	bfloat16	4,096
model.layers.13.self_attn.v_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.13.self_attn.v_proj.bias	Linear	512	bfloat16	512
model.layers.14	GptOssDecoderLayer	–	–	–
model.layers.14.input_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.14.mlp	GptOssMLP	–	–	–
model.layers.14.mlp.experts.gate_up_proj	GptOssExperts	32 × 2880 × 5760	bfloat16	530,841,600
model.layers.14.mlp.experts.gate_up_proj_bias	GptOssExperts	32 × 5760	bfloat16	184,320
model.layers.14.mlp.experts.down_proj	GptOssExperts	32 × 2880 × 2880	bfloat16	265,420,800
model.layers.14.mlp.experts.down_proj_bias	GptOssExperts	32 × 2880	bfloat16	92,160
model.layers.14.mlp.router.weight	GptOssTopKRouter	32 × 2880	bfloat16	92,160
model.layers.14.mlp.router.bias	GptOssTopKRouter	32	bfloat16	32
model.layers.14.post_attention_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.14.self_attn.sinks	GptOssAttention	64	bfloat16	64
model.layers.14.self_attn.k_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.14.self_attn.k_proj.bias	Linear	512	bfloat16	512
model.layers.14.self_attn.o_proj.weight	Linear	2880 × 4096	bfloat16	11,796,480
model.layers.14.self_attn.o_proj.bias	Linear	2880	bfloat16	2,880
model.layers.14.self_attn.q_proj.weight	Linear	4096 × 2880	bfloat16	11,796,480
model.layers.14.self_attn.q_proj.bias	Linear	4096	bfloat16	4,096
model.layers.14.self_attn.v_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.14.self_attn.v_proj.bias	Linear	512	bfloat16	512
model.layers.15	GptOssDecoderLayer	–	–	–
model.layers.15.input_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.15.mlp	GptOssMLP	–	–	–
model.layers.15.mlp.experts.gate_up_proj	GptOssExperts	32 × 2880 × 5760	bfloat16	530,841,600
model.layers.15.mlp.experts.gate_up_proj_bias	GptOssExperts	32 × 5760	bfloat16	184,320
model.layers.15.mlp.experts.down_proj	GptOssExperts	32 × 2880 × 2880	bfloat16	265,420,800
model.layers.15.mlp.experts.down_proj_bias	GptOssExperts	32 × 2880	bfloat16	92,160
model.layers.15.mlp.router.weight	GptOssTopKRouter	32 × 2880	bfloat16	92,160
model.layers.15.mlp.router.bias	GptOssTopKRouter	32	bfloat16	32
model.layers.15.post_attention_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.15.self_attn.sinks	GptOssAttention	64	bfloat16	64
model.layers.15.self_attn.k_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.15.self_attn.k_proj.bias	Linear	512	bfloat16	512
model.layers.15.self_attn.o_proj.weight	Linear	2880 × 4096	bfloat16	11,796,480
model.layers.15.self_attn.o_proj.bias	Linear	2880	bfloat16	2,880
model.layers.15.self_attn.q_proj.weight	Linear	4096 × 2880	bfloat16	11,796,480
model.layers.15.self_attn.q_proj.bias	Linear	4096	bfloat16	4,096
model.layers.15.self_attn.v_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.15.self_attn.v_proj.bias	Linear	512	bfloat16	512
model.layers.16	GptOssDecoderLayer	–	–	–
model.layers.16.input_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.16.mlp	GptOssMLP	–	–	–
model.layers.16.mlp.experts.gate_up_proj	GptOssExperts	32 × 2880 × 5760	bfloat16	530,841,600
model.layers.16.mlp.experts.gate_up_proj_bias	GptOssExperts	32 × 5760	bfloat16	184,320
model.layers.16.mlp.experts.down_proj	GptOssExperts	32 × 2880 × 2880	bfloat16	265,420,800
model.layers.16.mlp.experts.down_proj_bias	GptOssExperts	32 × 2880	bfloat16	92,160
model.layers.16.mlp.router.weight	GptOssTopKRouter	32 × 2880	bfloat16	92,160
model.layers.16.mlp.router.bias	GptOssTopKRouter	32	bfloat16	32

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Layer Path	Type	Weight Shape	Dtype	Params
model.layers.16.post_attention_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.16.self_attn.sinks	GptOssAttention	64	bfloat16	64
model.layers.16.self_attn.k_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.16.self_attn.k_proj.bias	Linear	512	bfloat16	512
model.layers.16.self_attn.o_proj.weight	Linear	2880 × 4096	bfloat16	11,796,480
model.layers.16.self_attn.o_proj.bias	Linear	2880	bfloat16	2,880
model.layers.16.self_attn.q_proj.weight	Linear	4096 × 2880	bfloat16	11,796,480
model.layers.16.self_attn.q_proj.bias	Linear	4096	bfloat16	4,096
model.layers.16.self_attn.v_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.16.self_attn.v_proj.bias	Linear	512	bfloat16	512
model.layers.17	GptOssDecoderLayer	–	–	–
model.layers.17.input_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.17.mlp	GptOssMLP	–	–	–
model.layers.17.mlp.experts.gate_up_proj	GptOssExperts	32 × 2880 × 5760	bfloat16	530,841,600
model.layers.17.mlp.experts.gate_up_proj_bias	GptOssExperts	32 × 5760	bfloat16	184,320
model.layers.17.mlp.experts.down_proj	GptOssExperts	32 × 2880 × 2880	bfloat16	265,420,800
model.layers.17.mlp.experts.down_proj_bias	GptOssExperts	32 × 2880	bfloat16	92,160
model.layers.17.mlp.router.weight	GptOssTopKRouter	32 × 2880	bfloat16	92,160
model.layers.17.mlp.router.bias	GptOssTopKRouter	32	bfloat16	32
model.layers.17.post_attention_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.17.self_attn.sinks	GptOssAttention	64	bfloat16	64
model.layers.17.self_attn.k_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.17.self_attn.k_proj.bias	Linear	512	bfloat16	512
model.layers.17.self_attn.o_proj.weight	Linear	2880 × 4096	bfloat16	11,796,480
model.layers.17.self_attn.o_proj.bias	Linear	2880	bfloat16	2,880
model.layers.17.self_attn.q_proj.weight	Linear	4096 × 2880	bfloat16	11,796,480
model.layers.17.self_attn.q_proj.bias	Linear	4096	bfloat16	4,096
model.layers.17.self_attn.v_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.17.self_attn.v_proj.bias	Linear	512	bfloat16	512
model.layers.18	GptOssDecoderLayer	–	–	–
model.layers.18.input_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.18.mlp	GptOssMLP	–	–	–
model.layers.18.mlp.experts.gate_up_proj	GptOssExperts	32 × 2880 × 5760	bfloat16	530,841,600
model.layers.18.mlp.experts.gate_up_proj_bias	GptOssExperts	32 × 5760	bfloat16	184,320
model.layers.18.mlp.experts.down_proj	GptOssExperts	32 × 2880 × 2880	bfloat16	265,420,800
model.layers.18.mlp.experts.down_proj_bias	GptOssExperts	32 × 2880	bfloat16	92,160
model.layers.18.mlp.router.weight	GptOssTopKRouter	32 × 2880	bfloat16	92,160
model.layers.18.mlp.router.bias	GptOssTopKRouter	32	bfloat16	32
model.layers.18.post_attention_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.18.self_attn.sinks	GptOssAttention	64	bfloat16	64
model.layers.18.self_attn.k_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.18.self_attn.k_proj.bias	Linear	512	bfloat16	512
model.layers.18.self_attn.o_proj.weight	Linear	2880 × 4096	bfloat16	11,796,480
model.layers.18.self_attn.o_proj.bias	Linear	2880	bfloat16	2,880
model.layers.18.self_attn.q_proj.weight	Linear	4096 × 2880	bfloat16	11,796,480
model.layers.18.self_attn.q_proj.bias	Linear	4096	bfloat16	4,096
model.layers.18.self_attn.v_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.18.self_attn.v_proj.bias	Linear	512	bfloat16	512
model.layers.19	GptOssDecoderLayer	–	–	–
model.layers.19.input_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.19.mlp	GptOssMLP	–	–	–
model.layers.19.mlp.experts.gate_up_proj	GptOssExperts	32 × 2880 × 5760	bfloat16	530,841,600
model.layers.19.mlp.experts.gate_up_proj_bias	GptOssExperts	32 × 5760	bfloat16	184,320
model.layers.19.mlp.experts.down_proj	GptOssExperts	32 × 2880 × 2880	bfloat16	265,420,800
model.layers.19.mlp.experts.down_proj_bias	GptOssExperts	32 × 2880	bfloat16	92,160
model.layers.19.mlp.router.weight	GptOssTopKRouter	32 × 2880	bfloat16	92,160
model.layers.19.mlp.router.bias	GptOssTopKRouter	32	bfloat16	32
model.layers.19.post_attention_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880

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Layer Path	Type	Weight Shape	Dtype	Params
model.layers.19.self_attn.sinks	GptOssAttention	64	bfloat16	64
model.layers.19.self_attn.k_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.19.self_attn.k_proj.bias	Linear	512	bfloat16	512
model.layers.19.self_attn.o_proj.weight	Linear	2880 × 4096	bfloat16	11,796,480
model.layers.19.self_attn.o_proj.bias	Linear	2880	bfloat16	2,880
model.layers.19.self_attn.q_proj.weight	Linear	4096 × 2880	bfloat16	11,796,480
model.layers.19.self_attn.q_proj.bias	Linear	4096	bfloat16	4,096
model.layers.19.self_attn.v_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.19.self_attn.v_proj.bias	Linear	512	bfloat16	512
model.layers.20	GptOssDecoderLayer	–	–	–
model.layers.20.input_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.20.mlp	GptOssMLP	–	–	–
model.layers.20.mlp.experts.gate_up_proj	GptOssExperts	32 × 2880 × 5760	bfloat16	530,841,600
model.layers.20.mlp.experts.gate_up_proj_bias	GptOssExperts	32 × 5760	bfloat16	184,320
model.layers.20.mlp.experts.down_proj	GptOssExperts	32 × 2880 × 2880	bfloat16	265,420,800
model.layers.20.mlp.experts.down_proj_bias	GptOssExperts	32 × 2880	bfloat16	92,160
model.layers.20.mlp.router.weight	GptOssTopKRouter	32 × 2880	bfloat16	92,160
model.layers.20.mlp.router.bias	GptOssTopKRouter	32	bfloat16	32
model.layers.20.post_attention_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.20.self_attn.sinks	GptOssAttention	64	bfloat16	64
model.layers.20.self_attn.k_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.20.self_attn.k_proj.bias	Linear	512	bfloat16	512
model.layers.20.self_attn.o_proj.weight	Linear	2880 × 4096	bfloat16	11,796,480
model.layers.20.self_attn.o_proj.bias	Linear	2880	bfloat16	2,880
model.layers.20.self_attn.q_proj.weight	Linear	4096 × 2880	bfloat16	11,796,480
model.layers.20.self_attn.q_proj.bias	Linear	4096	bfloat16	4,096
model.layers.20.self_attn.v_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.20.self_attn.v_proj.bias	Linear	512	bfloat16	512
model.layers.21	GptOssDecoderLayer	–	–	–
model.layers.21.input_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.21.mlp	GptOssMLP	–	–	–
model.layers.21.mlp.experts.gate_up_proj	GptOssExperts	32 × 2880 × 5760	bfloat16	530,841,600
model.layers.21.mlp.experts.gate_up_proj_bias	GptOssExperts	32 × 5760	bfloat16	184,320
model.layers.21.mlp.experts.down_proj	GptOssExperts	32 × 2880 × 2880	bfloat16	265,420,800
model.layers.21.mlp.experts.down_proj_bias	GptOssExperts	32 × 2880	bfloat16	92,160
model.layers.21.mlp.router.weight	GptOssTopKRouter	32 × 2880	bfloat16	92,160
model.layers.21.mlp.router.bias	GptOssTopKRouter	32	bfloat16	32
model.layers.21.post_attention_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.21.self_attn.sinks	GptOssAttention	64	bfloat16	64
model.layers.21.self_attn.k_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.21.self_attn.k_proj.bias	Linear	512	bfloat16	512
model.layers.21.self_attn.o_proj.weight	Linear	2880 × 4096	bfloat16	11,796,480
model.layers.21.self_attn.o_proj.bias	Linear	2880	bfloat16	2,880
model.layers.21.self_attn.q_proj.weight	Linear	4096 × 2880	bfloat16	11,796,480
model.layers.21.self_attn.q_proj.bias	Linear	4096	bfloat16	4,096
model.layers.21.self_attn.v_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.21.self_attn.v_proj.bias	Linear	512	bfloat16	512
model.layers.22	GptOssDecoderLayer	–	–	–
model.layers.22.input_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.22.mlp	GptOssMLP	–	–	–
model.layers.22.mlp.experts.gate_up_proj	GptOssExperts	32 × 2880 × 5760	bfloat16	530,841,600
model.layers.22.mlp.experts.gate_up_proj_bias	GptOssExperts	32 × 5760	bfloat16	184,320
model.layers.22.mlp.experts.down_proj	GptOssExperts	32 × 2880 × 2880	bfloat16	265,420,800
model.layers.22.mlp.experts.down_proj_bias	GptOssExperts	32 × 2880	bfloat16	92,160
model.layers.22.mlp.router.weight	GptOssTopKRouter	32 × 2880	bfloat16	92,160
model.layers.22.mlp.router.bias	GptOssTopKRouter	32	bfloat16	32
model.layers.22.post_attention_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.22.self_attn.sinks	GptOssAttention	64	bfloat16	64

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Layer Path	Type	Weight Shape	Dtype	Params
model.layers.22.self_attn.k_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.22.self_attn.k_proj.bias	Linear	512	bfloat16	512
model.layers.22.self_attn.o_proj.weight	Linear	2880 × 4096	bfloat16	11,796,480
model.layers.22.self_attn.o_proj.bias	Linear	2880	bfloat16	2,880
model.layers.22.self_attn.q_proj.weight	Linear	4096 × 2880	bfloat16	11,796,480
model.layers.22.self_attn.q_proj.bias	Linear	4096	bfloat16	4,096
model.layers.22.self_attn.v_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.22.self_attn.v_proj.bias	Linear	512	bfloat16	512
model.layers.23	GptOssDecoderLayer	–	–	–
model.layers.23.input_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.23.mlp	GptOssMLP	–	–	–
model.layers.23.mlp.experts.gate_up_proj	GptOssExperts	32 × 2880 × 5760	bfloat16	530,841,600
model.layers.23.mlp.experts.gate_up_proj_bias	GptOssExperts	32 × 5760	bfloat16	184,320
model.layers.23.mlp.experts.down_proj	GptOssExperts	32 × 2880 × 2880	bfloat16	265,420,800
model.layers.23.mlp.experts.down_proj_bias	GptOssExperts	32 × 2880	bfloat16	92,160
model.layers.23.mlp.router.weight	GptOssTopKRouter	32 × 2880	bfloat16	92,160
model.layers.23.mlp.router.bias	GptOssTopKRouter	32	bfloat16	32
model.layers.23.post_attention_layernorm.weight	GptOssRMSNorm	2880	bfloat16	2,880
model.layers.23.self_attn.sinks	GptOssAttention	64	bfloat16	64
model.layers.23.self_attn.k_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.23.self_attn.k_proj.bias	Linear	512	bfloat16	512
model.layers.23.self_attn.o_proj.weight	Linear	2880 × 4096	bfloat16	11,796,480
model.layers.23.self_attn.o_proj.bias	Linear	2880	bfloat16	2,880
model.layers.23.self_attn.q_proj.weight	Linear	4096 × 2880	bfloat16	11,796,480
model.layers.23.self_attn.q_proj.bias	Linear	4096	bfloat16	4,096
model.layers.23.self_attn.v_proj.weight	Linear	512 × 2880	bfloat16	1,474,560
model.layers.23.self_attn.v_proj.bias	Linear	512	bfloat16	512
lm_head.weight	Linear	201088 × 2880	bfloat16	579,133,440
model	GptOssModel	–	–	–
model.embed_tokens.weight	Embedding	201088 × 2880	bfloat16	579,133,440
model.layers	ModuleList	–	–	–
model.norm.weight	GptOssRMSNorm	2880	bfloat16	2,880

4 Parameter Count Estimation

Component	Parameters	% of Total
MLP	19,119,145,728	91.4%
Attention	637,203,456	3.0%
Embedding	579,133,440	2.8%
LM Head	579,133,440	2.8%
LayerNorm	141,120	0.0%
Total	**20,914,757,184**	**100.0%**

4.1 Per-Layer Parameter Count

Layer	Params	Running Total
model.layers.0.input_layernorm	2,880	2,880
model.layers.0.mlp.experts	796,538,880	796,541,760
model.layers.0.mlp.router	92,192	796,633,952
model.layers.0.post_attention_layernorm	2,880	796,636,832
model.layers.0.self_attn	64	796,636,896
model.layers.0.self_attn.k_proj	1,475,072	798,111,968
model.layers.0.self_attn.o_proj	11,799,360	809,911,328
model.layers.0.self_attn.q_proj	11,800,576	821,711,904
model.layers.0.self_attn.v_proj	1,475,072	823,186,976
model.layers.1.input_layernorm	2,880	823,189,856
model.layers.1.mlp.experts	796,538,880	1,619,728,736
model.layers.1.mlp.router	92,192	1,619,820,928
model.layers.1.post_attention_layernorm	2,880	1,619,823,808
model.layers.1.self_attn	64	1,619,823,872
model.layers.1.self_attn.k_proj	1,475,072	1,621,298,944
model.layers.1.self_attn.o_proj	11,799,360	1,633,098,304
model.layers.1.self_attn.q_proj	11,800,576	1,644,898,880
model.layers.1.self_attn.v_proj	1,475,072	1,646,373,952
model.layers.2.input_layernorm	2,880	1,646,376,832
model.layers.2.mlp.experts	796,538,880	2,442,915,712
model.layers.2.mlp.router	92,192	2,443,007,904
model.layers.2.post_attention_layernorm	2,880	2,443,010,784
model.layers.2.self_attn	64	2,443,010,848
model.layers.2.self_attn.k_proj	1,475,072	2,444,485,920
model.layers.2.self_attn.o_proj	11,799,360	2,456,285,280
model.layers.2.self_attn.q_proj	11,800,576	2,468,085,856
model.layers.2.self_attn.v_proj	1,475,072	2,469,560,928
model.layers.3.input_layernorm	2,880	2,469,563,808
model.layers.3.mlp.experts	796,538,880	3,266,102,688
model.layers.3.mlp.router	92,192	3,266,194,880
model.layers.3.post_attention_layernorm	2,880	3,266,197,760
model.layers.3.self_attn	64	3,266,197,824
model.layers.3.self_attn.k_proj	1,475,072	3,267,672,896
model.layers.3.self_attn.o_proj	11,799,360	3,279,472,256
model.layers.3.self_attn.q_proj	11,800,576	3,291,272,832
model.layers.3.self_attn.v_proj	1,475,072	3,292,747,904
model.layers.4.input_layernorm	2,880	3,292,750,784
model.layers.4.mlp.experts	796,538,880	4,089,289,664
model.layers.4.mlp.router	92,192	4,089,381,856
model.layers.4.post_attention_layernorm	2,880	4,089,384,736
model.layers.4.self_attn	64	4,089,384,800
model.layers.4.self_attn.k_proj	1,475,072	4,090,859,872
model.layers.4.self_attn.o_proj	11,799,360	4,102,659,232
model.layers.4.self_attn.q_proj	11,800,576	4,114,459,808
model.layers.4.self_attn.v_proj	1,475,072	4,115,934,880
model.layers.5.input_layernorm	2,880	4,115,937,760
model.layers.5.mlp.experts	796,538,880	4,912,476,640
model.layers.5.mlp.router	92,192	4,912,568,832
model.layers.5.post_attention_layernorm	2,880	4,912,571,712
model.layers.5.self_attn	64	4,912,571,776
model.layers.5.self_attn.k_proj	1,475,072	4,914,046,848
model.layers.5.self_attn.o_proj	11,799,360	4,925,846,208
model.layers.5.self_attn.q_proj	11,800,576	4,937,646,784
model.layers.5.self_attn.v_proj	1,475,072	4,939,121,856
model.layers.6.input_layernorm	2,880	4,939,124,736
model.layers.6.mlp.experts	796,538,880	5,735,663,616
model.layers.6.mlp.router	92,192	5,735,755,808
model.layers.6.post_attention_layernorm	2,880	5,735,758,688
model.layers.6.self_attn	64	5,735,758,752
model.layers.6.self_attn.k_proj	1,475,072	5,737,233,824
model.layers.6.self_attn.o_proj	11,799,360	5,749,033,184
model.layers.6.self_attn.q_proj	11,800,576	5,760,833,760